

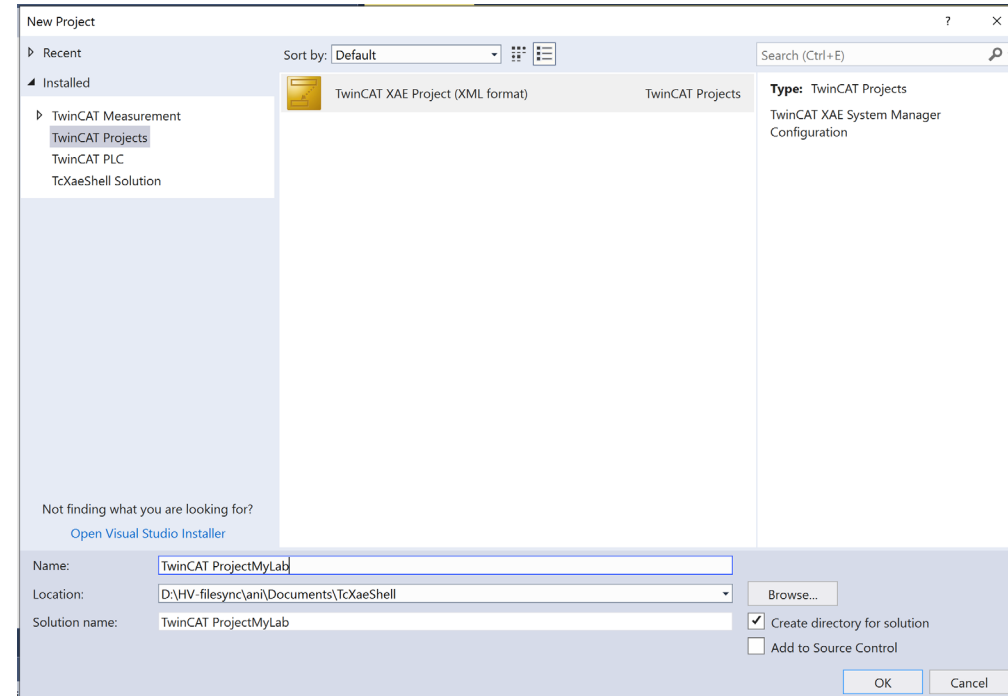
Lecture 10

TwinCAT Project from scratch and Visualisation

- Create a TwinCAT XAE project
- Create PLCs
- Connect Variables
- Visualization in TwinCAT

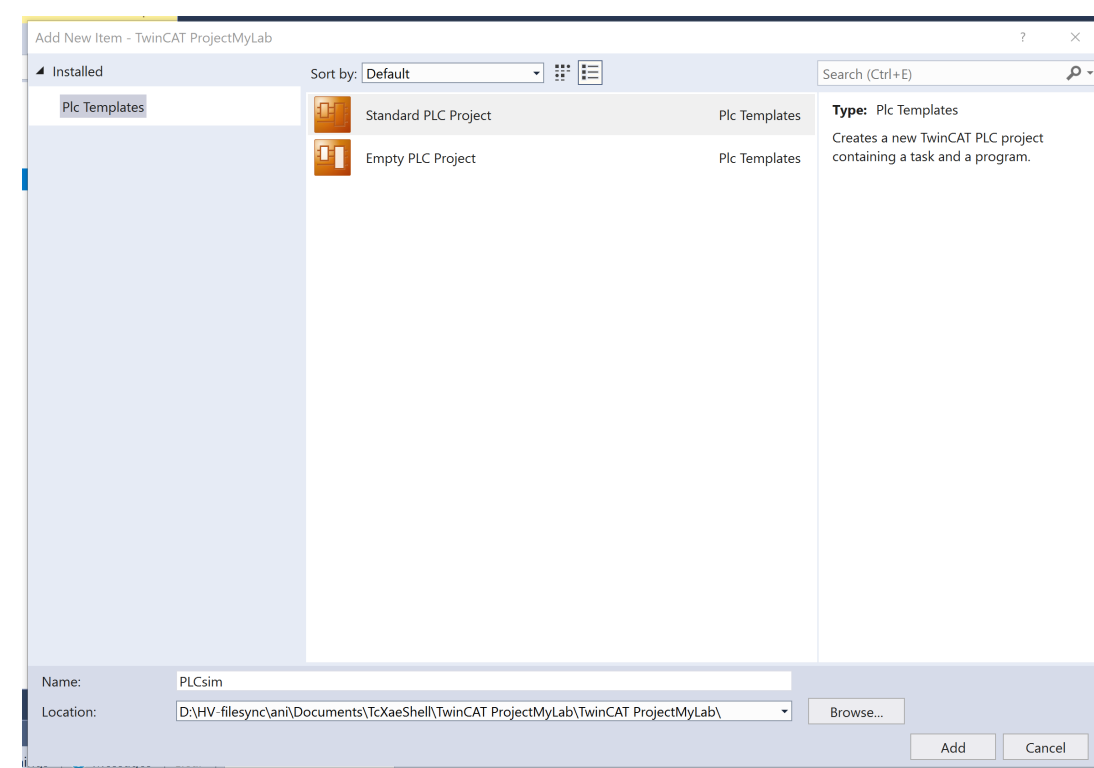
Create a TwinCAT XAE Solution

- Click on TwinCAT XAE Shell
- File New Project
- Give the project a good name and a good location
- The solution will get the same name as the project automatically
- Click on OK



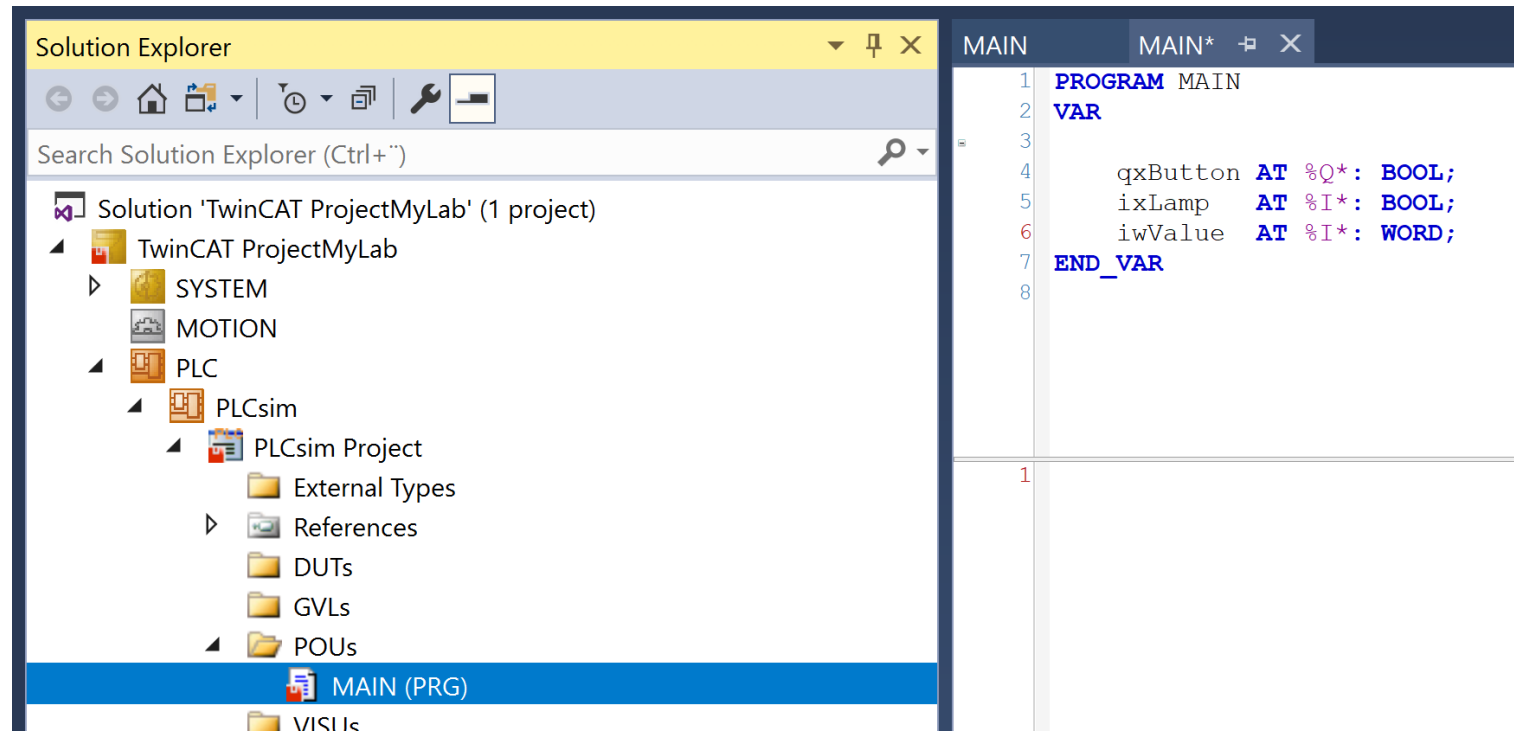
Add PLCs in TwinCAT

- We will add two PLCs to the solution
 - PLCsim
 - PLCstudent
- Right-click on PLC in the solution explorer
 - Add new Item
- Name PLCsim
- Select Standard PLC Project
- Click on the Add button
- Add PLCstudent in the same way



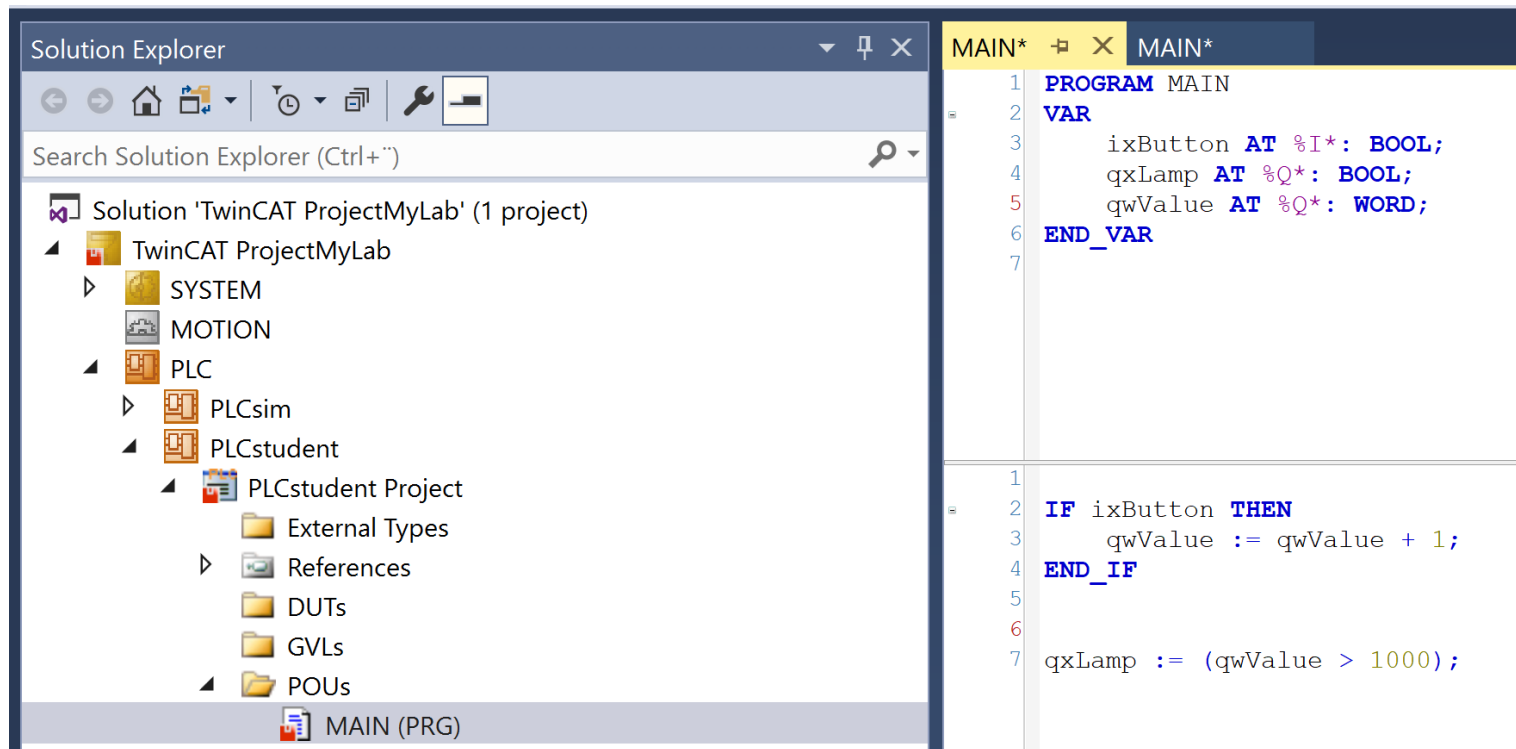
Simple demo example three variables

- Declare the variables in PLCsim as inputs %I* and outputs %Q*



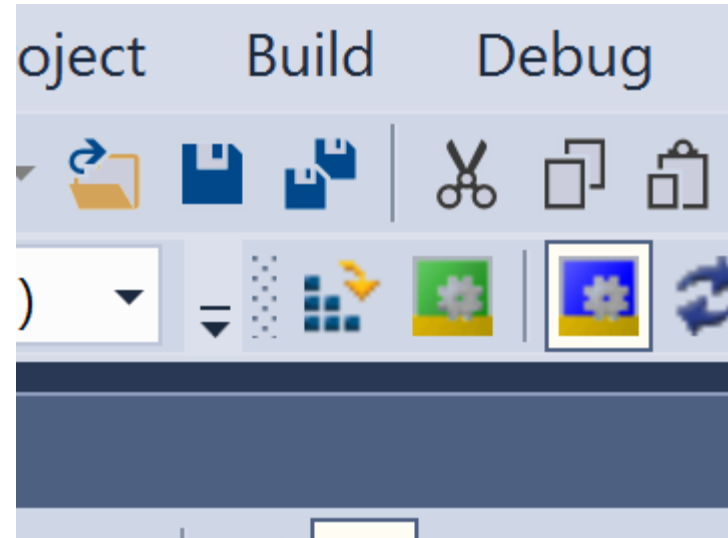
Create the corresponding variables in PLCstudent

- OBS! Inputs in PLCsim become Outputs in PLCstudent and vises verse
- Add a simple test program in MAIN



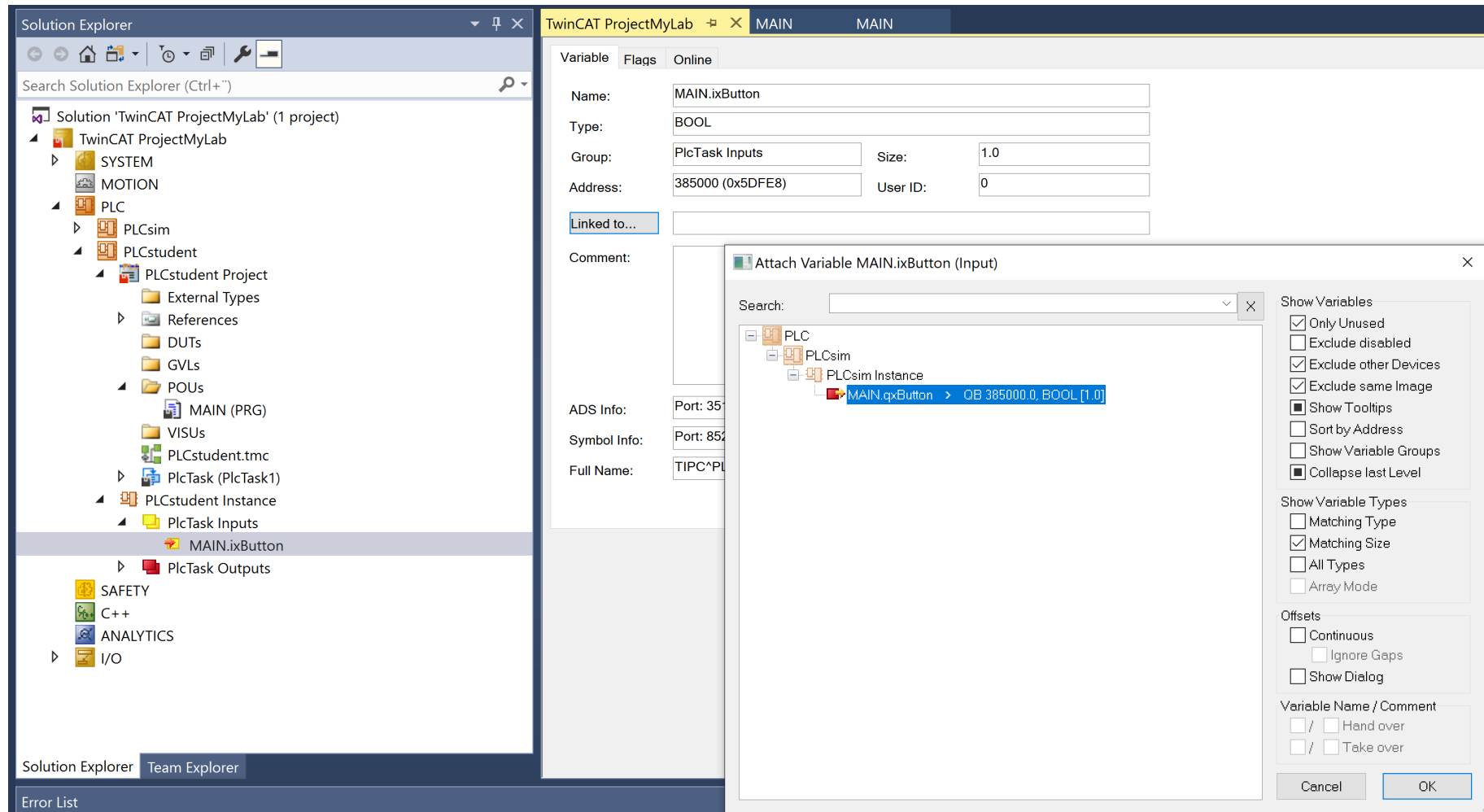
Connect the variables between PLCsim and PLCstudent

- Save all
- Reload all
- Activate Configuration
- Click OK for run mode

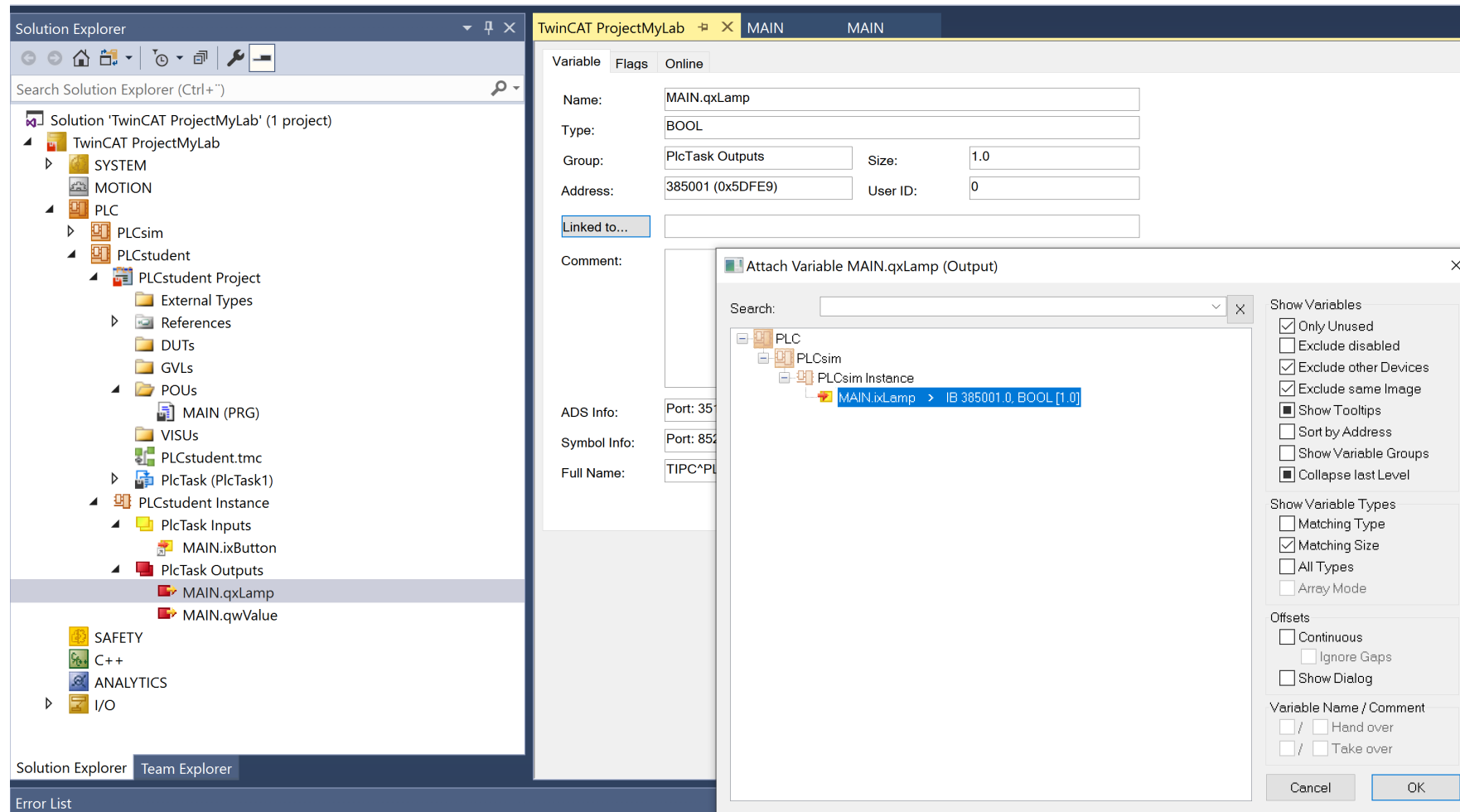


Link the PLCstudent input variable ixButton to PLCsim qxButton

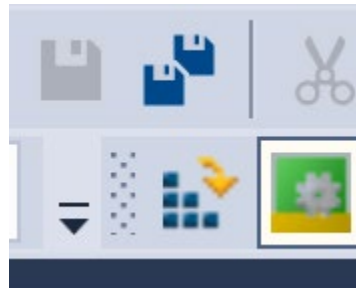
Dubbel-click on the variable and the Linked to button, select corresponding variable, and click OK



Link the PLCstudent output variables to corresponding PLCsim input variables

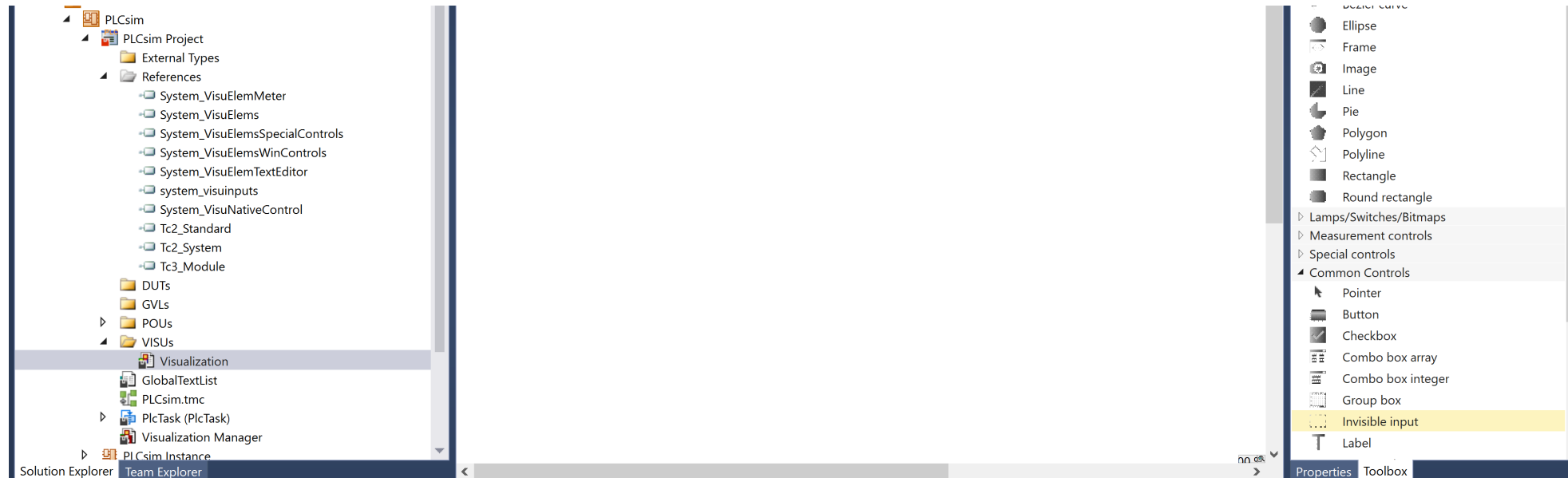


Save all and Activate Configuration

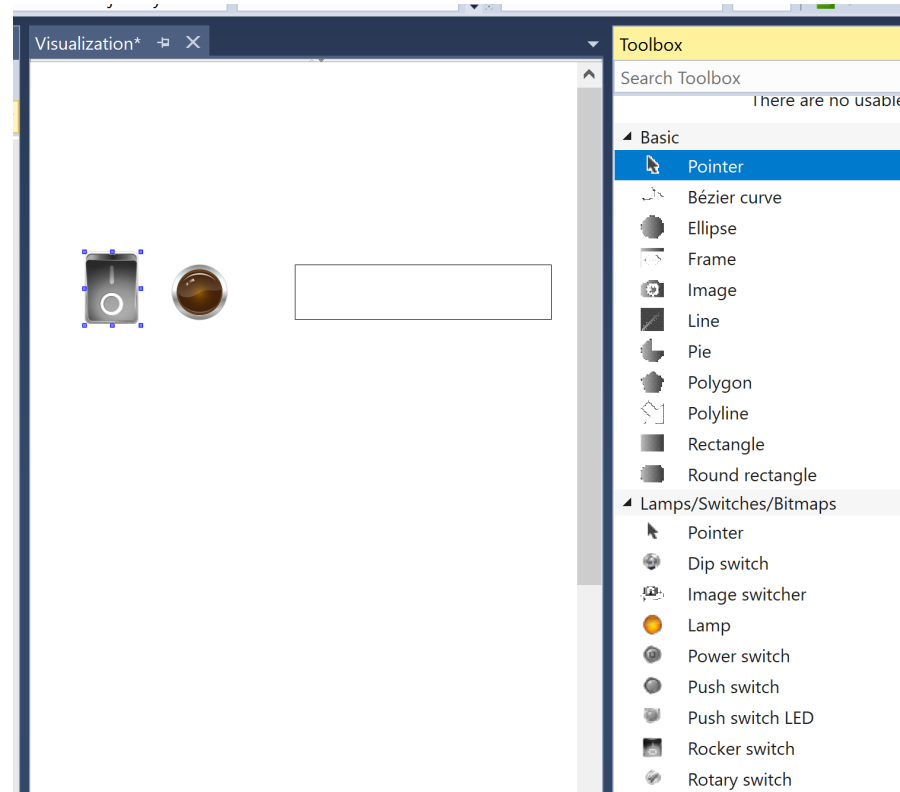


Create a visualization in PLCsim

- Right-click on VISUs, Select add visualization and Click Open
- If you not have a toolbox selected
 - Go to View and select toolbox



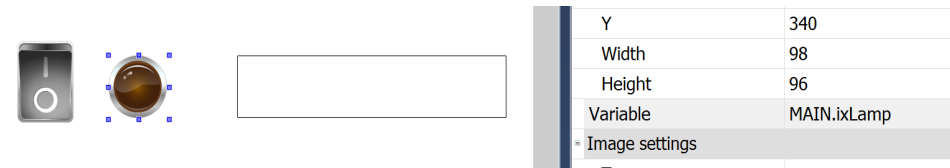
Add a rocker switch, lamp and Rectangle to the visualization



Properties

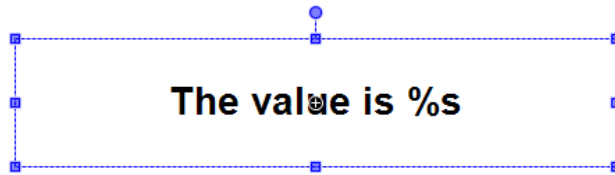
- Connect the variables to the lamp and the rocker switch

- Select Properties
- Select the rocker switch
- Write MAIN. In the Variable field
- Set Element behavior to Image toggler
 - Tapper is a push-button function
- Select qxButton
- Select the lamp
- Write MAIN. In the Variable field
- Select ixLamp



Text in the rectangle

- OBS! Small %s to have the variable value in the text



Texts	
Text	The value is %s
Tooltip	
Text properties	
Absolute movement	
Relative movement	
Text variables	
Text variable	MAIN.iwValue
Tooltip variable	
Dynamic texts	

Time to test rub

- Save all
- Activate configuration
- Log in and Start the PLCsim and PLCstudent
- Put the rocker switch on see the value increase and the lamp turns on after > 1000



The value is 7284

Play around

- A variable on position properties will let the object move
- There are a lot of other objects, Measurement controls
- It is also possible to create your own objects by creating Active X elements (Not included in this course)

The Visualization in TwinCAT

- Was from start created as a simple debug tool
- Has now been increased but still not a full SCADA
- Possible to run the visualization on an
 - Operator Panel
 - (Must have the same runtime version in the target system as in the engineering system)
 - Web server in the runtime system,
 - Let you surf into the visualization by a web browser!

Operator Panels (OP-Panels)

- Separate HMI software for the OP-Panel
 - A large number of graphical objects
 - Easy to use
- Communicates through variables (TAGs)
 - Pricing is per tags
- Communication
 - OPC
 - OPC-UA
 - Modbus
 - Fieldbuses
 - Or special brand-dependent communication



SCADA- Supervisory Control and Data Acquisition

- Computer-based systems that monitor, control and visualize a process Usually consists of
 - HMI
 - Computer for high-level control
 - Database for
 - Historical data
 - Recipes
 - Production plans
 - Communication for remote operation
 - OPC
 - OPC-UA
 - Modbus

